CAPSTONE, PORTFOLIO & INTERVIEWS · LESSON 3 OF 10

Data Scientist, ML and GenAI Interview Preparation: Questions and Model Answers

A compact question bank with model answers for data scientist, ML engineer and GenAI roles: Python and pandas, statistics, ML fundamentals, model evaluation, feature engineering, system and case design (churn, fraud, recommendations), MLOps, LLMs, RAG and agents, plus how to handle take-home assignments and live coding.

Career track · 2026 · Advanced · 120 min

By the end of this lesson you can

- Answer core Python, statistics and ML questions concisely
- Walk through an end-to-end ML system design
- Explain LLM, RAG and agent concepts and trade-offs
- Plan and present take-home assignments

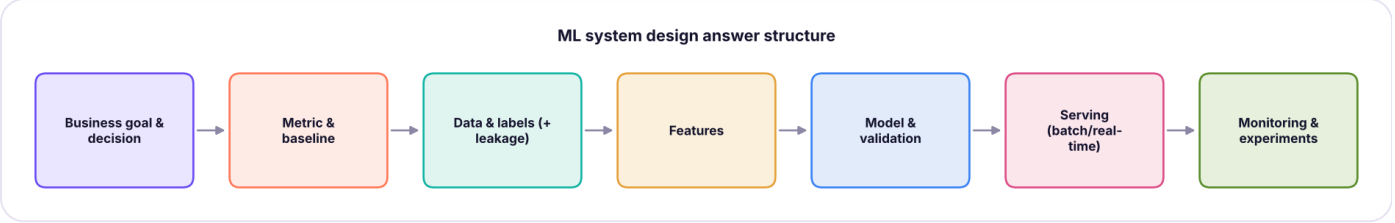
Python and pandas (quick answers)

Question	Answer
List vs tuple vs set vs dict?	Mutable sequence / immutable record / unique unordered / key→value; sets and dicts give O(1) lookups
`apply` vs vectorisation?	Vectorised column operations run in C and are much faster; avoid `apply(axis=1)` on large data
`merge` vs `join` vs `concat`?	merge on columns, join on index, concat stacks
How to handle missing values?	Understand why, then drop, impute (median, group-wise, model) or flag; impute inside CV
Generator vs list?	Generators are lazy and memory-efficient, consumed once

Statistics and ML fundamentals

Question	Model answer
Bias-variance trade-off?	Simple models underfit (bias), complex models overfit (variance); choose complexity by validation error
L1 vs L2?	L1 zeroes coefficients (selection); L2 shrinks smoothly; Elastic Net mixes
Why cross-validation?	A single split is noisy; k-fold averages and supports tuning without touching the test set
Precision vs recall?	Precision: flagged that are right; recall: positives caught; choose by error costs
ROC-AUC vs PR-AUC?	PR-AUC is more informative for rare positives
Bagging vs boosting?	Parallel averaging (variance) vs sequential error correction (bias)
How does gradient descent work?	Iteratively step weights against the loss gradient with a learning rate
What is data leakage?	Using information unavailable at prediction time; prevent with pipelines, time-based splits and feature cut-offs
p-value?	Probability of data at least this extreme if H0 were true; not the probability H0 is true

ML system design: a template



Answer ML design questions in this order.

Example: **"Design a fraud detection system for UPI payments."** Real-time decision under 100 ms; label = confirmed fraud (delayed, imbalanced); features: amount vs user history, device/location change, merchant risk, velocity, graph links; model: gradient boosting + rules + anomaly scores; evaluate PR-AUC and recall at alert capacity; serve with a feature store and low-latency API; monitor drift and attack patterns; human review queue; A/B test thresholds.

LLM, RAG and agent questions

Question	Model answer
How do LLMs work?	Transformers trained to predict the next token, then instruction-tuned and aligned with feedback
RAG vs fine-tuning?	RAG for changing knowledge with citations; fine-tuning for behaviour/format
How to evaluate a RAG system?	Retrieval hit rate/MRR + answer correctness, faithfulness, abstention on a labelled set
How to reduce hallucinations?	Grounding, tools, "say I don't know", citations, validation, evaluation
What is an agent?	An LLM in a loop choosing tool actions toward a goal, with memory and stopping conditions
What is MCP?	Open protocol to connect assistants to tools/data via servers (tools, resources, prompts)
Prompt injection defence?	Separate data from instructions, least-privilege tools, approvals, output checks
Cost control?	Model routing, caching, batching, shorter context and outputs

Take-home assignments

1. Restate the problem and assumptions in the first lines of your report.
2. Do a focused EDA; do not chart everything.
3. Baseline first, then 1–2 better models; validate correctly.
4. Translate results into business impact and a recommendation.
5. Clean, runnable code (requirements, README); a 5-slide summary.
6. State limitations and next steps honestly.

Live coding tips

Clarify inputs/outputs and edge cases, start with a simple correct solution, test with a small example, then optimise and state Big-O (Python DSA lessons 26–28).

KEY TAKEAWAYS

- Answer concisely, then go deeper only if asked.
- Use the design template: goal → metric → data → features → model → serving → monitoring.
- Know LLM/RAG/agent trade-offs, evaluation and safety.
- Take-homes: baseline, correct validation, business recommendation, clean code, honest limitations.

Quick quiz

4 questions · tap an answer to check it

1. What should you state first in an ML system-design answer?

A The business goal and the decision the model supports

B The algorithm

C The GPU type

D The library version

Answer: A. Framing first.

2. Best metric for a fraud model with 0.5% fraud?

A PR-AUC and recall at the alert capacity

B Accuracy

C R^2

D MAE

Answer: A. Rare positives.

3. RAG or fine-tuning for policy documents that change monthly?

- **A** RAG
- **B** Fine-tuning
- **C** Neither
- **D** Training from scratch

Answer: A. Knowledge belongs in retrieval.

4. What must a take-home report include besides the model?

- **A** Business impact, recommendation and limitations
- **B** Only code
- **C** Only accuracy
- **D** Every chart possible

Answer: A. Decision-makers read conclusions.

EXERCISE **Practice**

(1) Answer "Design a product recommendation system for an e-commerce home page" using the template. (2) Explain RAG to a non-technical manager in 60 seconds. (3) Do one timed take-home on the telecom churn data (3 hours).

Show solution (worked answer)

Recommendations: goal = revenue per session; candidates via collaborative filtering/embeddings + popularity; ranking model with user, item and context features; offline metrics (recall@k, NDCG) + online A/B test on revenue; cold start with popular/contextual items; monitor diversity and drift.

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